

# DEEP REINFORCEMENT LEARNING

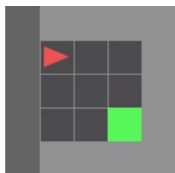
## 365.382 (SS26)

### Assignment I: Key and Door with DQN



LIT AI LAB  
Institute for Machine Learning

# MinigridEmpty5x5ImgObs Environment

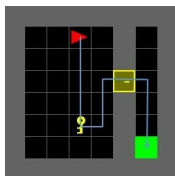


[Minigrid repository](#)

Easy environment  
training takes <10m  
**good for debugging**

- **Observation space:** Symbolic (non-image) observation (3, 7, 7)
- **Action space:** 7 discrete actions
  - **0:** left, **1:** right, **2:** forward
  - **3:** pick up, **4:** drop
  - **5:** use, **6:** done
- **Rewards:**  $1 - 0.9 \frac{\text{steps}}{\text{max steps}}$

# MinigridDoorKey6x6ImgObs Environment (Graded Task)



[Minigrid Webpage](#)

More difficult  
training takes 50m  
**graded task**

- **Observation space:** Symbolic (non-image) observation (3, 7, 7)
- **Action space:** 7 discrete actions
  - ☐ **0:** left , **1:** right, **2:** forward
  - ☐ **3:** pick up, **4:** drop
  - ☐ **5:** use, **6:** done
- **Reward Range:** {0, 1}

# Submission

- Submission server: <https://apps.ml.jku.at/challenge>
- Export your trained agent as ONNX file
- Upload ONNX file to submission server
  - ☐ Code submission closes on **Friday, May 15th, 18:00.**
  - ☐ Max model file size: **50MB.**
- Submit your **code**, **best model** and a short **report** to Moodle
  - ☐ Report the performance on your local setting.
  - ☐ Moodle submission stays open 24h after the challenge closes.
- Please upload a zip file named `k<studentid>.zip` containing:
  - ☐ `code.ipynb` / `code.py`
  - ☐ `model.onnx`
  - ☐ `report.pdf`

# Evaluation

■  $< 0.5$  : 0 pts

■  $\geq 0.5$  : 2 pts

■  $\geq 0.7$  : 4 pts

■  $\geq 0.8$  : 6 pts

■  $\geq 0.9$  : 8 pts

# Presentations

- If you are willing to give a short presentation of your work, fill out the poll Model that closes Sunday evening after the submission deadline.
- Students selected to present, will receive +2 bonus points.